

Jia et al. 2024 — MetaboAgeMort

Technical Assessment

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Context

These questions were framed in the context of the Biolearn Meeting assessment of metabolomic aging clocks. This technical report evaluates the Jia et al. 2024 paper (MetaboAgeMort) along the same dimensions previously used for Zhang et al. 2024 and Mutz et al. 2024.

1. Is the Clock NMR-Based? Which Specific Platform?

Yes. MetaboAgeMort uses Nightingale Health NMR spectroscopy, identical to Zhang 2024 and Mutz 2024. This is the same high-throughput ¹H-NMR automated platform producing 249 standard biomarkers as deployed in the UK Biobank. The platform dependency is an explicit limitation acknowledged in the paper.

Platform	Technology	Biomarkers	Notes
Nightingale Health ★	¹ H-NMR automated	249 standard	UK Biobank standard; used by Jia, Zhang, and Mutz 2024
Bruker IVDr	¹ H-NMR	~40 lipoproteins	Different quantification algorithms; not cross-compatible
LipoProfile (LabCorp)	NMR	20–30 lipoproteins	Clinical; different subclass definitions

2. Which Molecules Are Measured? What Are the 35 Key Biomarkers?

The model starts from 249 Nightingale NMR biomarkers (168 absolute concentrations + 81 ratios), expands to 325 total with 76 researcher-computed ratios. From 185 mortality-associated candidates identified by adjusted Cox regression ($P < 0.05/249$), LASSO Cox penalization with 10-fold cross-validation selects the final 35 metabolic biomarkers plus chronological age.

Category	Count	Key Biomarkers
Amino Acids	6	Alanine, histidine, leucine, valine, phenylalanine, tyrosine
Ketone Bodies	4	3-hydroxybutyrate, acetate, acetoacetate, acetone
Fatty Acids / PUFA	4	Linoleic acid, omega-3/total FA ratio, MUFA/total FA ratio, linoleic acid/total FA ratio
Lipoproteins	15+	VLDL avg. diameter, TG in very large VLDL, free cholesterol in very large HDL, total lipids in small HDL, CE in small HDL, TG/TL in large VLDL, PL/TL in very small VLDL, PL/TL in IDL, CE/TL in IDL, TG/TL in large LDL, TG/TL in medium LDL, PL/TL in small LDL, FC/TL in small LDL, CE/TL in very large HDL, FC/TL in small HDL
Glycolysis	3	Glucose, pyruvate, citrate
Inflammation	1	Glycoprotein acetyls (GlycA)
Fluid Balance	2	Creatinine, albumin

Note: The 35-biomarker composition closely overlaps with Zhang 2024's 54 biomarkers for the inflammation, fluid balance, and glycolysis categories. Jia 2024 explicitly demonstrates a stronger mortality signal than MetaboAge (which uses chronological age as a training target) but comparable or superior to MetaboHealth (which also targets mortality).

3. Platform Specificity Requirements

MetaboAgeMort inherits the same platform-specificity concerns as Zhang 2024. The paper notes that metabolomic profiles are sensitive to both NMR platform algorithms and sample processing conditions. The 15+ lipoprotein ratio biomarkers are the most platform-sensitive components.

- Processing problem: Different NMR platforms use proprietary deconvolution algorithms, yielding different numerical values from the same physical sample
- Lipoprotein subclasses: Defined and calibrated differently across platforms; direct cross-platform substitution is not valid
- Amino acids & glycolysis markers: More consistent across NMR platforms than lipoprotein ratios, but still platform-calibrated

Unlike Zhang 2024, Jia 2024 builds on concentration units rather than cohort-scaled values, which the authors argue improves cross-cohort applicability within the same Nightingale platform. This does not, however, resolve cross-platform incompatibility.

4. Sample Type Requirements

The model was trained on EDTA plasma samples from the UK Biobank. The paper does not explicitly report serum or heparin plasma validation, mirroring the limitation noted for Zhang 2024.

Sample Type	Status	Key Difference
EDTA Plasma ★	Trained on this	Retains clotting factors; EDTA chelates calcium; recommended sample type
Serum	Not validated	Clotting releases amino acids and alters metabolite levels; use with caution
Heparin Plasma	Not validated	Different anticoagulant effects; compatibility unknown

5. Implementation Availability

The implementation situation for Jia 2024 is partially more transparent than Zhang 2024, but remains restricted.

Item	Available?	Notes
Model coefficients (Gompertz)	Partial	Published in paper (Table S4) — unlike Zhang 2024 which requires contacting authors
LASSO selection lambda	Yes	10-fold cross-validation procedure described; reproducible with glmnet R package

Code / Software package	No	No standalone code repository or R package released
UK Biobank data access	Restricted	Requires application via ukbiobank.ac.uk; ~£4,000+ academic base fee; 2–3 month timeline
GWAS summary statistics	Yes	99 genomic loci published; available via supplementary data
External validation data	No	Only internal UK Biobank test split; no independent cohort validation reported

Implementation steps: (1) Obtain 249 Nightingale NMR biomarkers from EDTA plasma; (2) Apply concentration-based LASSO Cox model using published Gompertz coefficients (Table S4 of paper); (3) Compute MetaboAgeMortAccel as the residual from regressing MetaboAgeMort on chronological age.

6. Data Sources

- Primary: UK Biobank — 239,291 participants; median age 58.3 years; 53% female; 95.6% white European ethnicity
- Follow-up: Median 13.9 years; 20,447 deaths recorded
- Training/Test split: 70/30 random split — no independent external cohort used for validation
- Note: Absence of independent external validation is an explicit limitation of this study, contrasting with Zhang 2024 which validated in Generation Scotland (n=15,788)

7. Performance Summary

Metric	Value / Finding
Age correlation (r)	r = 0.85 overall (female: 0.85, male: 0.86)
MAE range	~4.78 years SD in MetaboAgeMortAccel; range −16.4 to +41.3 years acceleration
All-cause mortality HR	HR = 1.07 per year (95% CI: 1.06–1.08); robust across 8 subgroup stratifications
Cause-specific mortality	Cancer HR=1.06, CVD HR=1.08, Respiratory HR=1.08, Digestive HR=1.12, Neurodegenerative HR=1.03
C-statistic improvement	+0.017 over conventional risk factors (p < 0.001); IDI = 0.018
vs MetaboHealth (mortality)	Both maintain significance when mutually adjusted; MetaboAgeMort shows better AIC fit
vs MetaboAge (age-based)	MetaboAgeMort significantly higher AUC for 10-year mortality (p < 0.001)

8. Unique Contributions: Modifiable Factors & GWAS

This is the key differentiating contribution of Jia 2024 relative to all other metabolomic aging clocks.

Modifiable Factors (n=99):

- Pulmonary function — strongest category among 16 highest-impact factors
- Body composition metrics

- Socioeconomic status (Townsend deprivation index)
- Dietary quality scores
- Smoking status (current/former)
- Alcohol intake frequency
- Prevalent disease status

Genetic Analysis (GWAS):

- 99 genomic risk loci identified for MetaboAgeMort
- 271 genes mapped
- Tissue enrichment: liver (strongest signal)
- Distinct from Zhang 2024 GWAS which focused on individual biomarker-level loci (GCST90445833–GCST90446157)

9. Summary Assessment

Scientific Quality: Strong. Largest cohort size for mortality-trained metabolomic clock, rigorous LASSO-Cox methodology, comprehensive secondary analyses including GWAS and modifiable factor mapping. The use of concentration units rather than cohort-scaled values is methodologically superior for cross-cohort comparisons within the Nightingale platform.

Practical Concerns:

- Platform lock-in: Nightingale NMR only — same limitation as Zhang 2024
- Sample type: EDTA plasma required; serum compatibility unconfirmed
- No external validation: Internal test split only; independent cohort validation pending
- Code: No released package; Gompertz coefficients are in supplementary tables but require manual implementation
- UK Biobank access: ~£4,000+ and 2–3 months if retraining required

Recommendation for Biolearn:

PRIMARY recommendation when the research question includes understanding what drives metabolic aging or designing lifestyle/clinical interventions. The 99 modifiable factors and GWAS analyses are not available from any other NMR metabolomic clock and constitute a unique scientific contribution. For pure short-term mortality prediction performance, Zhang 2024 (AUC 86.8%) remains marginally superior. For chronological age estimation, Mutz 2024 (MileAge) should be preferred.

Reference: Jia, X. et al. A novel metabolomic aging clock predicting health outcomes and its genetic and modifiable factors. *Adv. Sci.* 11, 2406670 (2024). DOI: 10.1002/advs.202406670

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